

## 1 Thread Safety

Two threads touching the same collection at the same moment = corrupted data.

Thread-safe collections put a lock on every operation only one thread gets in at a time. Safer, but slower.

The default in Java is: fast and unsafe.

Only **Vector** and **Stack** are the collection that have Thread-Safe.

## 2 Vector vs ArrayList

Vector :

- Every method has synchronized on it.
- Even in single-thread code, you pay the lock cost every time.
- Grows by 100% when full wastes memory.

ArrayList :

- No locking at all — just does the work directly.
- Grows by 50% when full more efficient.
- Always your default choice for any list.
- Add thread safety yourself only if you need it.

## 3 ArrayDeque vs Stack

ArrayDeque:

- No index access. No inserting in the middle.
- Only push, pop, peek — exactly what a stack should be.
- Implements Deque — the correct interface..

Stack:

- Thread-safe, but inherits all of Vector's slowness too.

Extends Vector — wrong parent class .

- A stack should only allow push, pop, peek. Nothing else .